

# LOCALIZING FEATURES WITH MASKING FOR SATELLITE AND DEBRIS CLASSIFICATION

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## ABSTRACT

**In this work, we propose a localization and masking-based satellite and debris classification technique. SPACraft Recognition leveraging Knowledge of space environment (SPARK) dataset consists of 120K images where both RGB and corresponding Depth images are available. However, the depth images are noisy and inaccurate and significantly affect the classification task performance. To address this issue, we first create mask images of the RGB images which are used as input to the Convolutional Neural Network (CNN) for efficient classification of different satellites and debris. The depth images are first de-noised and hole filled using a simple morphological opening operation. Then masked images are calculated using both RGB and processed depth images. This masking operation provides two advantages: 1. it removes noise and fills the holes in the depth images and 2. it highlights satellites and debris while suppressing other information which does not contribute towards the classification task. We use the pre-trained EfficientNet B4 architecture and fine-tuned it with an edition of Global average pooling (GAP) and three dense layers. Our results show that the inclusion of the masking operation significantly improves the overall classification performance, achieving 97.76% accuracy on the validation data.**

## 1. INTRODUCTION

Satellites have transformed the daily life of communities - starting from the broadcast of television, transmission of telephone signals, navigation systems, weather forecasting, and so on. Considering the increasing dependency on satellites for several vital activities, space situational awareness (SSA) has become essential to identify and navigate around any surrounding objects including debris in space. Space debris is defunct man-made objects that orbit around the earth. Further, depending on the functional application, different kinds of satellites orbit around the earth at different heights and in different trajectories. Recognizing and classifying such satellites and debris using in space is a crucial task. Space agencies of different countries have planned missions to remove and capture space debris. Most of the existing techniques de-

pend on passive sensors for pose estimation and space object classification. However, this is a challenging task, as there is no reliable way to extract visual and scalable features of such targets [1].

Image classification is the task of assigning an image to one or more of a defined set of target classes. Different machine learning techniques have been explored by researchers across the globe for the image classifications task [2]. Image classification techniques can be adapted to the task of identifying the type of satellite or other categories of objects including debris. This is an upcoming and nascent area of research [1]. In the dawn of the deep learning era, the extraction of deep multi-scale features with the use of CNN models has opened up transfer learning-based solutions for space object classifications. Early work on space object classification was proposed by Linares and Furfaro [3], where a layered hierarchical deep CNN architecture was used to extract multi-scale features from brightness measurements and further used for space object classification. Furfaro *et al.* [4] reported another study of space object classification and characterization using a set of CNNs and Recurrent Neural Network (RNN) architectures using light-curve measurements. The study included the performance comparison of reflectance models and real light-curve data [4]. Afsar and Lu [5] reported the use of CNN architecture with transfer learning framework and data augmentation for satellite classification. Recently, Liu and Schreiber [6] developed a photometric observations-based space objects classification technique using a long-short term memory type deep learning architecture. Considering the importance of SSA application, Shen *et al.* [7] proposed a 3D-CNN model to study the evasive satellite behaviors from ground-based sensors.

While deep learning techniques are often data-hungry, the availability of training data in the space context is a challenge. There are two prominent datasets in this domain: the 6-D spacecraft pose estimation dataset (SPEED) [8] and the Unreal Rendered Spacecraft On-Orbit (URSO) [9] dataset. These datasets contain 15,000 images each. A new SPACraft Recognition leveraging Knowledge of Space environment (SPARK) [10] dataset was introduced as part of a data-driven challenge on SSA. It contains 150,000 RGB images with bounding box information and corresponding 150,000

depth images. These images belong to 11 class objects: 10 spacecraft and 1 debris.

In this paper, we describe our proposed system for SPARK challenge [11] of satellite and debris classification using the localization and masking process. In the first stage of the proposed scheme, we create the mask images, and then these masked images are used as input to the Convolutional Neural Network (CNN) for the efficient classification of different satellites and debris. In the proposed scheme, we adhered to a pre-trained EfficientNet B4 architecture and fine-tuned it with an addition of Global average pooling (GAP) and three dense layers. The masked images are calculated using both the RGB and depth images. Before application of the masked image, the depth images are denoised and the hole filled using morphological opening operation. The results of the proposed scheme show a significant improvement in the overall classification accuracy due to the inclusion of the masking operation. The proposed algorithm achieves 97.76% accuracy on the validation data.

## 2. PROPOSED WORK



**Fig. 1.** (a) Depth Image (b) Depth image after applying opening operation

Our proposed work localizes the object present in the given image by leveraging RGB and their corresponding depth images. However, the estimated depth images are noisy, inaccurate, and often contain holes. To address this issue, in the first step, we propose to improve the visual accuracy of depth images using a simple morphological opening operation. These modified depth images are further used to create masks to highlight the object of interest and suppress the information which is not helpful in the efficient classification. In the second step, the masked images are fed into a fine-tuned EfficientNet B4 architecture to predict the final 11 categories in the classification task. The details of each step are given below.

### 2.1. Morphological Operation

Most of the depth images in the SPARK dataset are not accurate and can not be directly used for creating masks. An example depth image from the SPARK dataset is shown in Fig.

1(a) and from this Fig, it can be observed that the depth image of the satellites contains inconsistencies around the object edges. We fill these inconsistencies by performing a morphological opening operation – erosion followed by dilation [12]. The erosion operation removes small noise components and dilation operation helps in filling the holes present in depth images.

The eroded image  $I_{e,o}$  is obtained by erosion operation on depth image  $I$  as:

$$I_{e,o} = I \ominus S = \{(m, n) : S_{m,n} \subset R_d\} \quad (1)$$

$S$  is the structuring element and  $S_{m,n}$  is the pixel  $(m, n)$  location on which the structural element is applied. In the proposed algorithm, we use a structuring element of size  $5 \times 5$ . The erosion operation is able to remove the small noise component and also shrink the size of the object. The dilation operation helps in increasing the size of the object and also fills the holes in the object. The dilated image  $D_o$  is obtained as:

$$D_o = E_{e,o} \oplus S = \{(m, n) : S_{m,n} \subset R_d\} \quad (2)$$

Here  $S$  is the structuring element that has an operation size similar to erosion -  $S$  is  $5 \times 5$ . Dilation operation is used for expanding shapes present in the image; it adds an extra layer of pixels to the outer and inner boundaries of the objects in the image.

Fig. 1(b) shows the depth image obtained using the morphological opening operation. It can be observed that a simple opening operation removes the noise and fills the holes. These modified depth images are then used in creating masks.

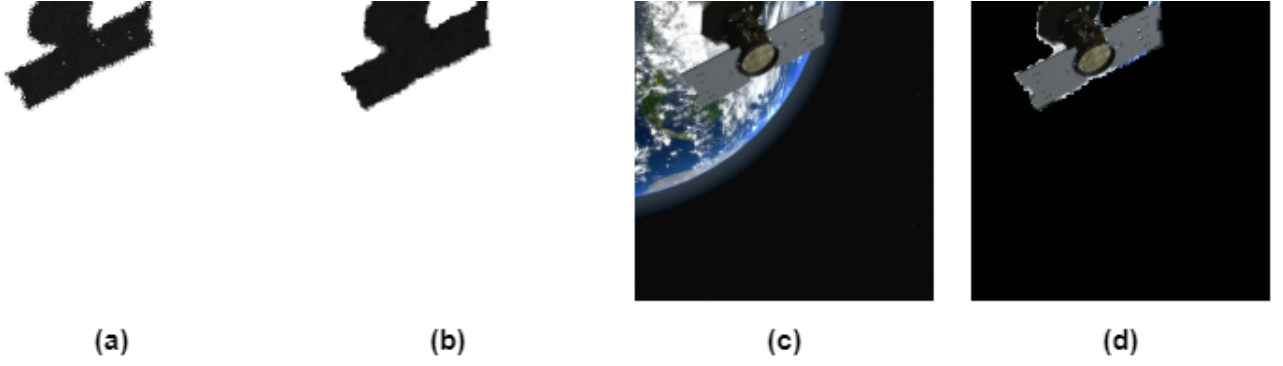
### 2.2. Masking

Since the objective is to classify satellites and debris, other information apart from satellite and debris available in the RGB image can potentially contribute to noise for the classification task. Further, the modified depth images have information about 10 different satellites and debris. We create masked images (as shown in Fig. 2(d)) which focus on the visual information pertinent to satellites and debris. The modified depth images estimated using equation (2) are grayscale and it is required to convert them into the binary image for the mask.

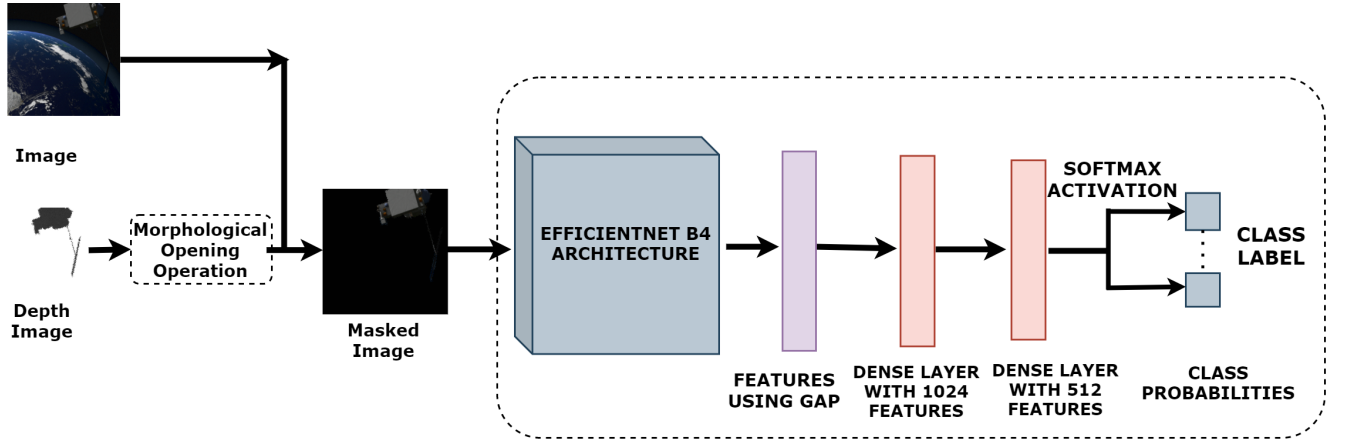
The mask image can be represented as:

$$M(i, j) = \begin{cases} 1, & \text{if } D_o \geq th \\ 0, & \text{else} \end{cases} \quad (3)$$

The masked images are estimated by simply taking the AND operation between the modified depth images and the RGB images. This step removes all the background objects and highlights only the object of interest in the output masked image. The mask image is represented as:



**Fig. 2.** (a) Depth Image (b) Depth image after applying opening operation (c) RGB image (d) Masked image



**Fig. 3.** An elaborated workflow of the proposed 11 class classification algorithm.

$$M_{masked} = I_{mg} \text{ AND } M \quad (4)$$

Here *AND* is the logical operator. In Fig. 2(d), we show the impact of masking and it can be observed that in the final masked image only object of interest is present and other unwanted information has been suppressed. In the proposed algorithm, we have selected the value of *th* to be 150. In the second step, these masked images are fed into the CNN architecture for the classification task.

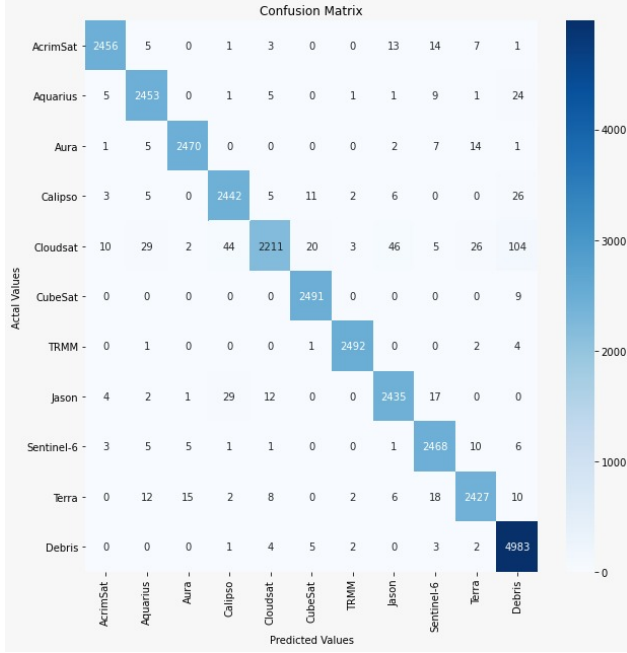
### 2.3. Network Architecture

After estimating the masked image, these images are fed in the recently proposed EfficientNet B4 architecture. EfficientNet B4 [13], a convolutional neural architecture that scales depth, width and resolution using uniform scaling. It is efficient and has been successfully used in many applications [14]. The pre-trained weights obtained from training EfficientNet B4 on ImageNet [15] dataset are used for initializing the network. After the last convolutional block, we added a global average pooling (GAP) layer and replaced the final fully connected (FC) network with three FC layers: the first

two layers have 1024 and 512 neurons and use the ReLU activation function. The output layer has 11 neurons with a softmax activation function, the output of which is the vector comprising of class probabilities. For training the model, we used categorical cross-entropy loss function, Adam optimizer, with default parameters and a learning rate of 0.0001. The implementation was based on Python 3.7.3 and Tensorflow 2.4.0. The architecture of the proposed algorithm is shown in Fig.3. The network parameters are summarized in Table 1. We also tested DenseNet [16], [17] as the baseline model.

**Table 1.** Network Architecture of the Proposed Algorithm

| Module                             | Input shape | Output shape |
|------------------------------------|-------------|--------------|
| EfficientNet-B4                    | (256,256,3) | (8,8,2048)   |
| Global Average Pooling             | (8,8,2048)  | (2048)       |
| Fully Connected Dense Layer + ReLU | (2048)      | (1024)       |
| Fully Connected Dense Layer + ReLU | (1024)      | (512)        |
| Fully Connected Dense Layer + ReLU | (512)       | (11)         |



**Fig. 4.** Confusion Matrix representing the Predicted class vs the Actual class.

### 3. RESULTS

The SPARK dataset has 90,000 training and 30,000 validation images with their corresponding depth images and we have shown the performance of the proposed algorithm (in terms of accuracy, precision, recall, and F1-score) on the validation dataset. We compared the performance of the proposed algorithm on classifying 11 categories using different features on the validation set in Table 2. We used DenseNet and EfficientNet architectures for the CNN part of the model. In the baseline approach, we use Densenet 121 [16], as the base architecture in the above-proposed architecture and obtained an accuracy of 84%. The inclusion of masking significantly improves the performance of the proposed algorithm. Also, EfficientNet B4 pre-trained network performs better than DenseNet and the proposed algorithm is able to achieve 0.979, 0.976, 0.977 values of precision, recall, and F1-score, respectively.

Fig. 4 shows the  $11 \times 11$  sized confusion matrix to validate whether the proposed algorithm is biased towards any particular class. Fig.5 shows the results in terms of precision, recall, F1-score for each of the 11 classes. From this figure, it can be seen that the proposed algorithm is giving satisfactory results on all of the 11 classes. From these results, it can be validated that the proposed algorithm is efficiently able to classify images into 11 categories.

**Table 2.** Results of the proposed method on SPARK validation set and the ablation study.

| Model                  | Accuracy |
|------------------------|----------|
| DenseNet-121           | 84%      |
| DenseNet-121 + Mask    | 93%      |
| EfficientNet B5 + Mask | 95%      |
| EfficientNet B4 + Mask | 97.76%   |

|            | precision | recall | f1-score |
|------------|-----------|--------|----------|
| AcrimSat   | 0.99      | 0.98   | 0.99     |
| Aquarius   | 0.97      | 0.98   | 0.98     |
| Aura       | 0.99      | 0.99   | 0.99     |
| Calipso    | 0.97      | 0.98   | 0.97     |
| Cloudsat   | 0.98      | 0.88   | 0.93     |
| CubeSat    | 0.99      | 1.00   | 0.99     |
| TRMM       | 1.00      | 1.00   | 1.00     |
| Jason      | 0.97      | 0.97   | 0.97     |
| Sentinel-6 | 0.97      | 0.99   | 0.98     |
| Terra      | 0.98      | 0.97   | 0.97     |
| Debris     | 0.96      | 1.00   | 0.98     |

**Fig. 5.** Performance of the proposed algorithm on individual 11 classes.

### 4. CONCLUSION

In this work, we proposed an efficient satellite and debris classification algorithm based on RGB and depth images. The first step of the proposed algorithm is to remove the noise and fill the holes in depth images using the simple morphological opening operation. Using the modified depth images, we created masks that highlight the object of interest and suppress the information which is not useful for the satellite/debris image classification. In the second step, the masked images are input to the EfficientNet B4 architecture for object classification. The experimental results validate the inclusion of masking in the proposed algorithm. The proposed algorithm achieves 97.76% accuracy on classifying the images into 11 categories on the validation data. Future works can explore the feasibility of co-object classification and detection simultaneously.

## 5. REFERENCES

- [1] Haoyue Zeng and Yong Xia, "Space target recognition based on deep learning," 07 2017, pp. 1–5.
- [2] Salihah Yousuf, "Analytical study of image classification using deep learning," *International Journal of Advanced Research in Computer Science*, vol. 9, pp. 754–759, 2018.
- [3] Richard Linares and Roberto Furfaro, "Space object classification using deep convolutional neural networks," in *2016 19th International Conference on Information Fusion (FUSION)*, 2016, pp. 1140–1146.
- [4] R. Furfaro, R. Linares, and V. Reddy, "Space objects classification and characterization via deep learning and light curves: Applications to space traffic management," in *2019 Progress through Collaboration-Space Traffic Management Conference*, 2019.
- [5] Roya Afshar and Shuai Lu, *Classification and Recognition of Space Debris and Its Pose Estimation Based on Deep Learning of CNNs*, pp. 605–613, 07 2020.
- [6] Tong Liu and K. Ulrich Schreiber, "Photometric space object classification via deep learning algorithms," *Acta Astronautica*, vol. 185, pp. 161–169, 2021.
- [7] Dan Shen, Carolyn Sheaff, Mengqing Guo, Erik Blasch, Khanh Pham, and Genshe Chen, "Three-dimensional convolutional neural network (3D-CNN) for satellite behavior discovery," in *Sensors and Systems for Space Applications XIV*, Genshe Chen and Khanh D. Pham, Eds. International Society for Optics and Photonics, 2021, vol. 11755, pp. 24 – 41, SPIE.
- [8] Mate Kisantal, Sumant Sharma, Tae Ha Park, Dario Izzo, Marcus Mörtens, and Simone D’Amico, "Satellite pose estimation challenge: Dataset, competition design, and results," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 4083–4098, 2020.
- [9] Pedro F Proença and Yang Gao, "Deep learning for spacecraft pose estimation from photorealistic rendering," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2020, pp. 6007–6013.
- [10] Mohamed Adel Musallam, Kassem Al Ismaeil, Oye-bade Oyedotun, Marcos Damian Perez, Michel Poucet, and Djamila Aouada, "Spark: Spacecraft recognition leveraging knowledge of space environment," *arXiv preprint arXiv:2104.05978*, 2021.
- [11] Mohamed Adel Musallam, Vincent Gaudilliere, Enjie Ghorbel, Kassem Al Ismaeil, Marcos Damian Perez, Michel Poucet, and Djamila Aouada, "Spacecraft recognition leveraging knowledge of space environment: Simulator, dataset, competition design, and analysis," *2021 IEEE International Conference on Image Processing, Grand Challenges*, 2021.
- [12] Vinit Jakhetiya, Ke Gu, Trisha Singhal, Sharath Chandra Guntuku, Zhifang Xia, and Weisi Lin, "A highly efficient blind image quality assessment metric of 3-d synthesized images using outlier detection," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 7, pp. 4120–4128, 2019.
- [13] Mingxing Tan and Quoc Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International Conference on Machine Learning*. PMLR, 2019, pp. 6105–6114.
- [14] Shubham Chaudhary, Sadbhawna Sadbhawna, Vinit Jakhetiya, Badri N Subudhi, Ujjwal Baid, and Sharath Chandra Guntuku, "Detecting covid-19 and community acquired pneumonia using chest ct scan images with deep learning," in *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 8583–8587.
- [15] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A Large-Scale Hierarchical Image Database," in *CVPR09*, 2009.
- [16] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4700–4708.
- [17] Palak Mahajan, Vinit Jakhetiya, Pawanesh Abrol, Parveen K. Lehana, Badri N. Subudhi, and Sharath C. Guntuku, "Perceptual quality evaluation of hazy natural images," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 12, pp. 8046–8056, 2021.